

K - Map Assignment

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I. ABSTRACT

This paper explains a Karnaugh maps (K-map) by finding the logic functions for the incrementing decoder from 0 to 9 and don't care condition using arduino uno.

II. COMPONENTS

The required components list is given in Table: I., seven segment display is shown in Fig.1, and 7447 IC pin diagram is shown in Fig-2.

Components	Value	Quantity
IC	7447	1
seven segment display		1
Arduino	UNO	1
Jumper Wires		50
Breadboard		1

TABLE I

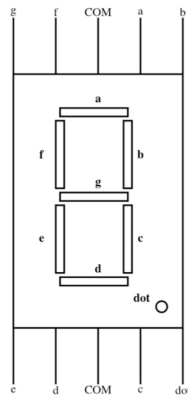


Fig. 1.

III. PROCEDURE

- 1) Make the connections of arduino, and 7447 ICs according to Fig-4.
- 2) Truth Table for k-map without don't care and incrementing from 0 to 9 :
- 3) Truth Table for k-map with don't care condition :
- 4) Execute the arduino code without any errors.
- 5) After upload the code into hardware setup using arduino IDE platform with hex file.



Fig. 2.

7447	$\bar{a}$	$\bar{b}$	$\bar{c}$	$\bar{d}$	$\bar{e}$	$\bar{f}$	$\bar{g}$
Display	a	b	c	d	e	f	g

Fig. 3.

Fig. 3.

IV. RESULTS

- 1) Download the code given in the link below and execute them to see the output as shown in Fig.6,7.
- 2) <https://github.com/madhureddy727/FWC-Assignments>

V. CONCLUSION

Hence implementation of K-Map using 7447 IC and Seven segment display using arduino UNO is done.

<i>Z</i>	<i>Y</i>	<i>X</i>	<i>W</i>	<i>D</i>	<i>C</i>	<i>B</i>	<i>A</i>
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0

TABLE II

<i>Z</i>	<i>Y</i>	<i>X</i>	<i>W</i>	<i>D</i>	<i>C</i>	<i>B</i>	<i>A</i>
0	0	0	0	1	0	0	1
0	0	0	1	0	0	0	0
0	0	1	0	0	0	0	1
0	0	1	1	0	0	1	0
0	1	0	0	0	0	1	1
0	1	0	1	0	1	0	0
0	1	1	0	0	1	0	1
0	1	1	1	0	1	1	0
1	0	0	0	0	1	1	1
1	0	0	1	1	0	0	0
1	0	1	0	-	-	-	-
1	0	1	1	-	-	-	-
1	1	0	0	-	-	-	-
1	1	0	1	-	-	-	-
1	1	1	0	-	-	-	-
1	1	1	1	-	-	-	-

TABLE III

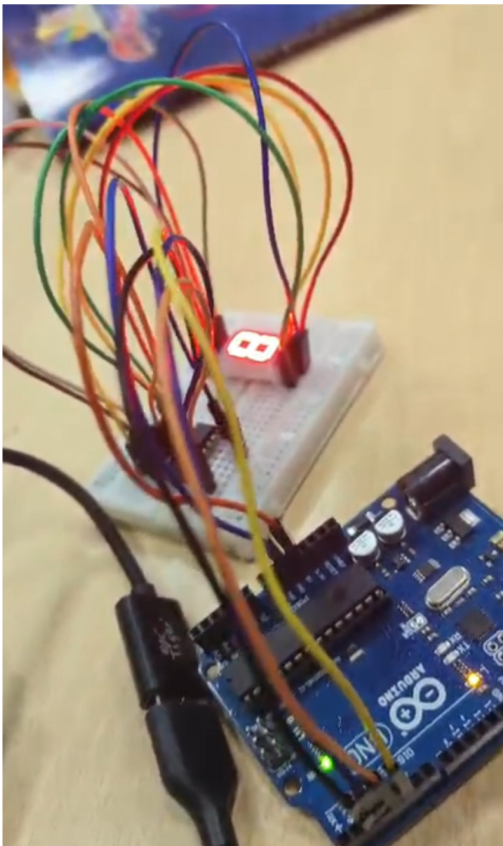


Fig. 4.